E-COMMERCE FUNNEL ANALYSIS

Dashboard Case Study — Power BI

DAX · Data Modeling · Funnel Analytics · Drill-through

Tool Power BI	**Dataset** 90,000 Users · 4 Funnel Stages
Pages 3 Main + 4 Drill-throughs	**Focus Areas** CVR · Drop-off · Segment · Time Trends

Prepared as part of Data Analytics Portfolio · 2026

1. Project Overview

This case study presents a comprehensive funnel analysis dashboard built in Power BI to evaluate user behavior across four key stages of an e-commerce digital journey. The dashboard identifies conversion bottlenecks, quantifies drop-offs at each stage, and delivers actionable business insights through interactive drill-through navigation.

Business Problem

E-commerce businesses routinely lose revenue at invisible points in their conversion funnel. Without stage-level visibility, optimization decisions are based on guesswork rather than evidence. Three core questions drive this project:

- Where do users abandon the journey, and at what rate?
- Which device or gender segment underperforms and why?
- Are drop-off patterns structural problems or time-bound anomalies?

End-to-End Funnel Results

90,000	0.50%	87%	452
Total Users	End-to-End CVR	Search Drop-off	Completed Journey

Stage Transition	Users In	Users Out	CVR	Drop-off %
Home → Search	90,000	45,000	**50%**	50%
Search → Payment	45,000	6,030	**13%**	87%
Payment → Confirm	6,030	452	**7%**	93%

Key Finding: *Search → Payment is the primary revenue leak — 87% of users are lost here. Even a 5% improvement at this stage would yield ~2,250 additional conversions with zero increase in traffic spend.*

Dataset Overview

The E-commerce Website Funnel Analysis Dataset (Kaggle) comprises five CSV tables tracking user-level interactions across each funnel stage, joined via user_id.

Table	Key Column	Description
home_page_table	user_id, page	Entry point — all users who landed on the home page
search_page_table	user_id, page	Users who progressed to the search/product discovery page
payment_page_table	user_id, page	Users who initiated a transaction at the payment page
payment_confirmation_table	user_id, page	Users who completed purchase — successful conversion
user_table	user_id, date, device, sex	Visit date, device type, gender — primary key for all joins

Data Modeling Approach

- **Funnel Restructuring (Wide → Long):** Page columns from all four stage tables were merged into user_table via user_id joins, then unpivoted into a Stage column and a binary Visited column (1 = visited, 0 = not visited).
- **Binary stage flags** — visited_home, visited_search, visited_payment, visited_confirm — created for clean conditional filtering.
- **Key DAX measures** — CVR, Drop-off %, Stage Users, Conversion Contribution %, Drop Rank, Drop Severity — all built as explicit measures.

2. Dashboard Pages

The dashboard comprises three main analytical pages and four drill-through pages, all wired through interactive slicers and stage-level navigation.

PAGE 1 — Funnel Overview

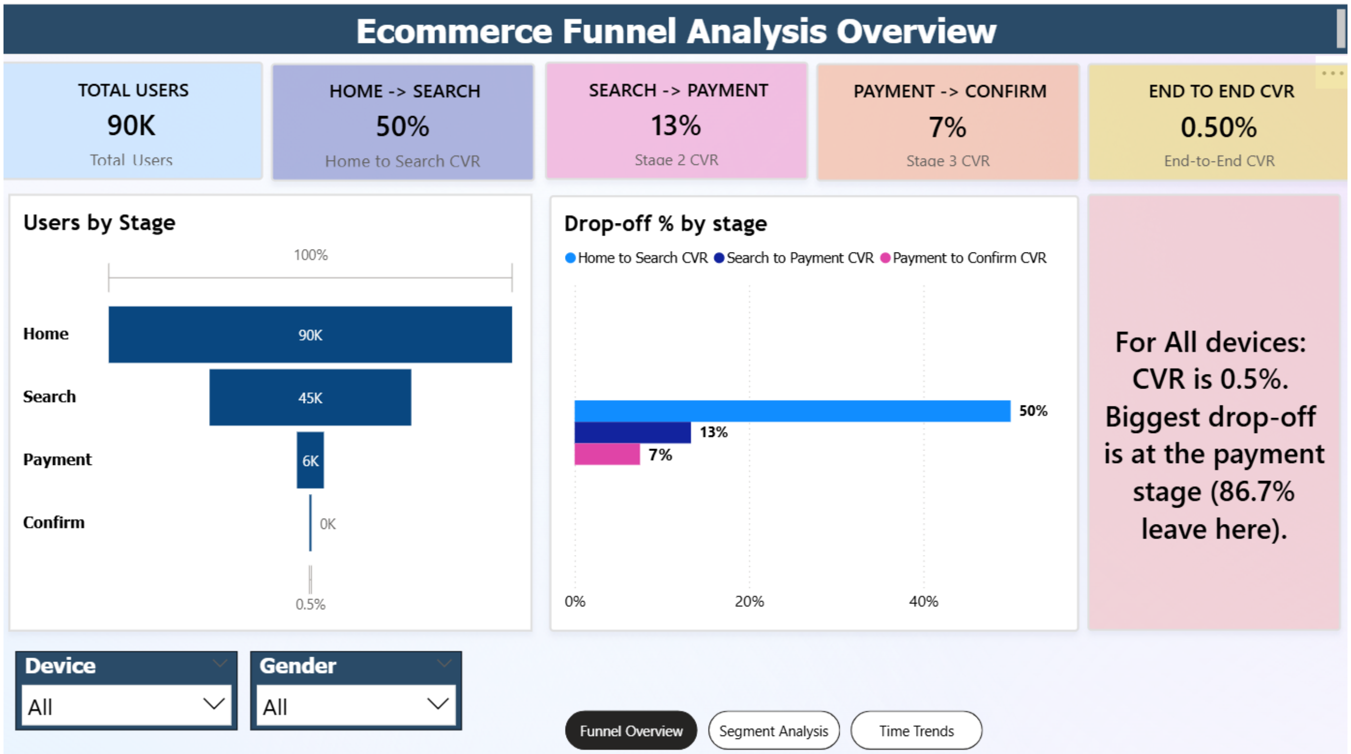
Funnel Overview

High-level snapshot of the entire funnel — quickly identify the biggest drop-off points

Purpose

The Overview page is the entry point for any stakeholder. At a glance it communicates total user volume, stage-by-stage conversion rates, and the most critical drop-off in the funnel — all before a single filter is touched.

KPI Cards	Five headline metrics at the top**: Total Users (90K) · Home→Search CVR (50%) · Search→Payment CVR (13%) · Payment→Confirm CVR (7%) · End-to-End CVR (0.50%).** Gives instant funnel health without reading a chart.
Users by Stage Bar	Horizontal bar chart showing the dramatic volume collapse: 90K → 45K → 6K → ~0. The visual cliff between Search and Payment makes the bottleneck unmissable.
Drop-off % Chart	Three-point line chart plotting CVR% across each transition. The steep dive from 50% to 13% at Search→Payment is immediately apparent and prompts the right question.
Dynamic Insight Card	A DAX-driven text card that auto-generates a sentence identifying the worst drop-off stage and device context based on active slicer selections — updates live.
Slicers	Device (All / Mobile / Desktop) and Gender (All / Female / Male) — applied globally across all visuals on the page for cross-segment comparison.



Finding: *Search → Payment loses 87% of users — the single largest revenue leak in the funnel. Fixing even 5% of this drop adds ~2,250 conversions at zero acquisition cost.*

PAGE 2 — Segment Analysis

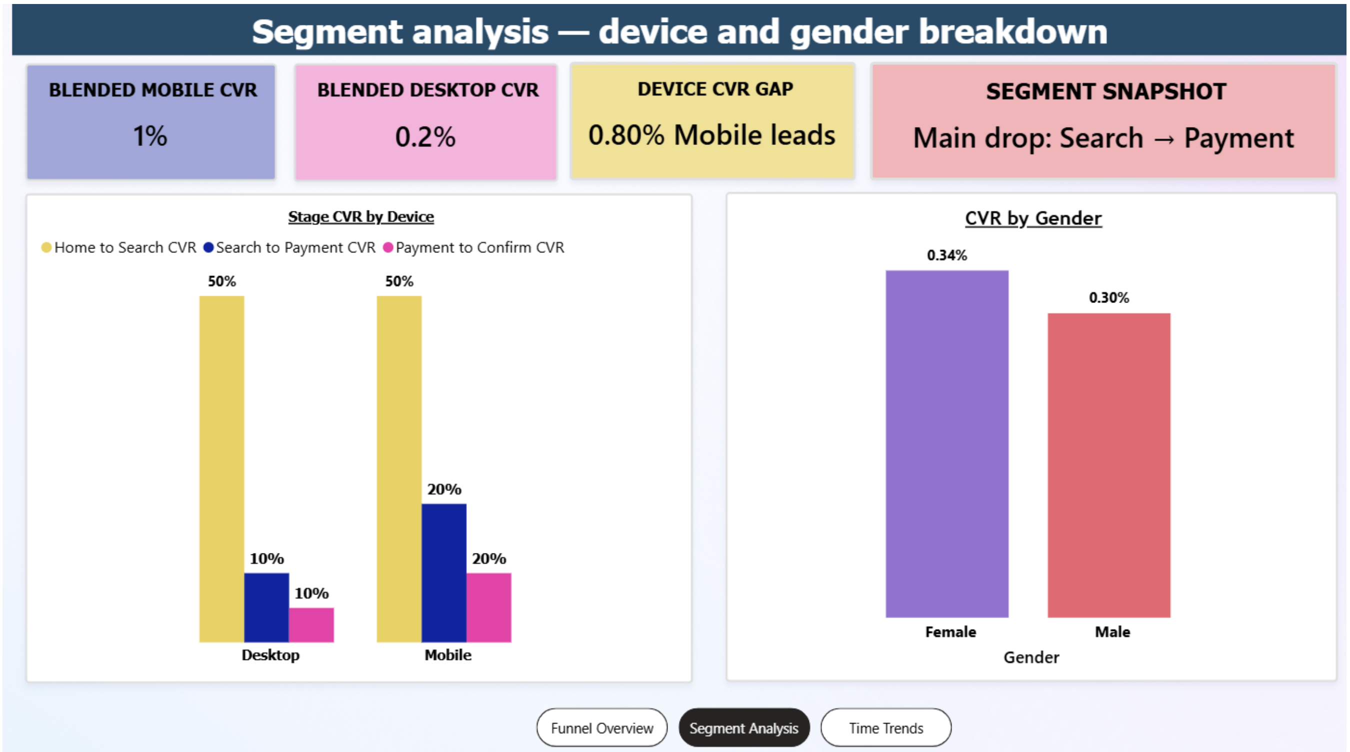
Segment Analysis

Break conversion down by device and gender — find who drives results and who doesn't

Purpose

The Segment page answers 'who converts?' rather than just 'how many convert?'. By splitting CVR across device type and gender, it isolates the highest-leverage intervention point — avoiding spray-and-pray optimization.

KPI Cards	Blended Mobile CVR (1%) · Blended Desktop CVR (0.2%) · Device CVR Gap (+0.80%, Mobile leads) · Segment Snapshot card that dynamically names the top-performing segment.
Stage CVR by Device	Clustered bar chart comparing Desktop vs Mobile CVR at each stage transition. Reveals exactly where device performance diverges, not just the blended rate.
CVR by Gender	Simple bar — Female (0.34%) vs Male (0.30%) end-to-end CVR. The near-identical gap confirms gender is not the lever; device experience is.
Conversion Contribution %	Desktop Female (33.39%) + Desktop Male (33.21%) = ~66% of all conversions despite ~50% of users. Desktop users convert at near-zero rate in absolute terms.



Finding: *Mobile users drive ~66% of conversions. Desktop CVR is near zero — the device experience gap is the dominant optimization lever, not gender.*

PAGE 3 — Time Trend Analysis

Time Trend Analysis

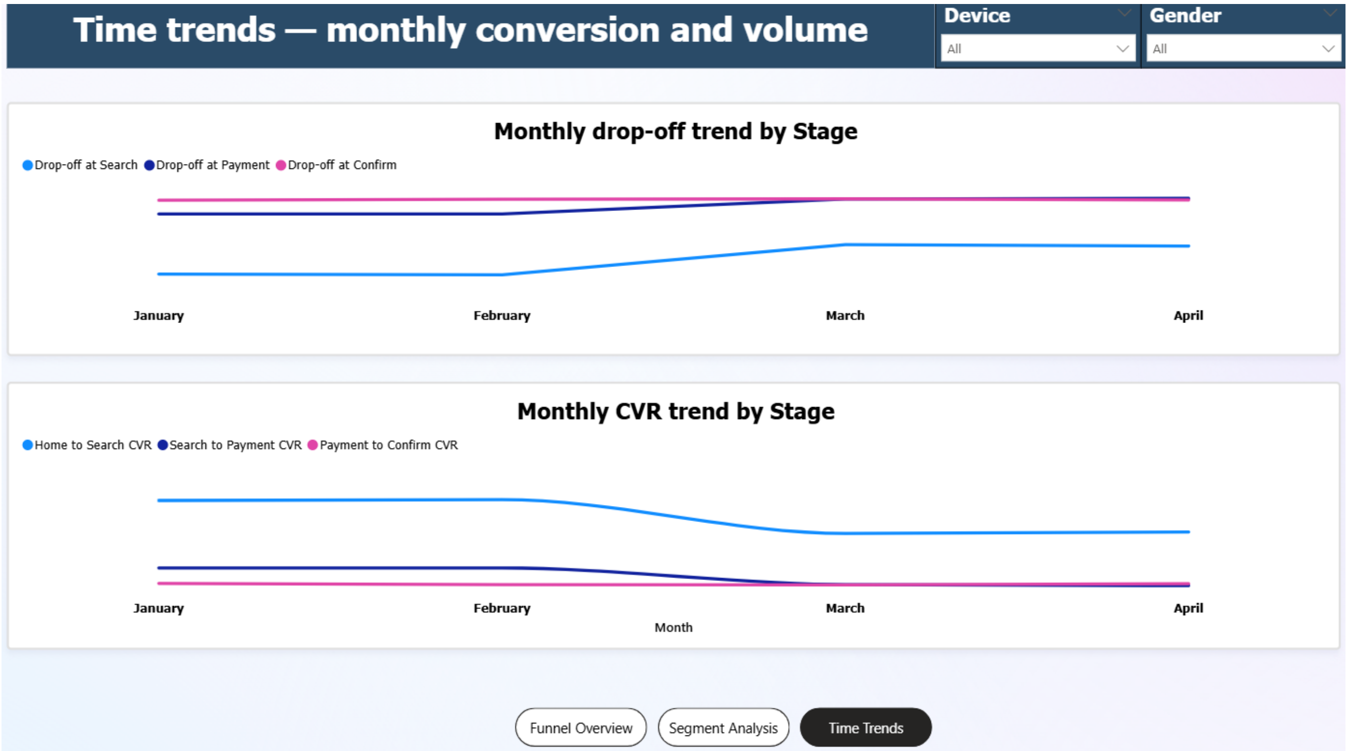
Track CVR and drop-off month-by-month — separate structural problems from time-bound anomalies

Purpose

The Trend page answers 'is this getting worse over time?' — a critical question before allocating engineering resources. It tracks January to April across all three key funnel transitions, filtered by device and gender.

Drop-off Trend Chart	Line chart overlaying monthly drop-off % at Search, Payment, and Confirm. Shows whether stages are worsening together (systemic) or independently (stage-specific bug).
CVR by Stage Trend	Tracks Home→Search, Search→Payment, Payment→Confirm CVR across four months. Declining Home→Search from January signals growing top-of-funnel friction.
Key Observations	Search drop-off is consistently highest. Payment + Confirm drop-off converges upward from March — a worsening late-stage problem. Daily CVR trends up Feb→Mar then plateaus in April.
Slicers	Device and Gender filters applied across both line charts — enabling segment-specific time-trend inspection side by side.

4 Months Tracked	**3** Stages in Trend	↑ Overall CVR Trend *Gradual — Feb to Mar*	⚠ Late-Stage Risk *Worsening from March*



Finding: *Payment and Confirm drop-off converges upward from March — a worsening late-stage problem. Home→Search CVR declining from January signals growing top-of-funnel friction.*

PAGE 4 — Stage Drill-Through Pages

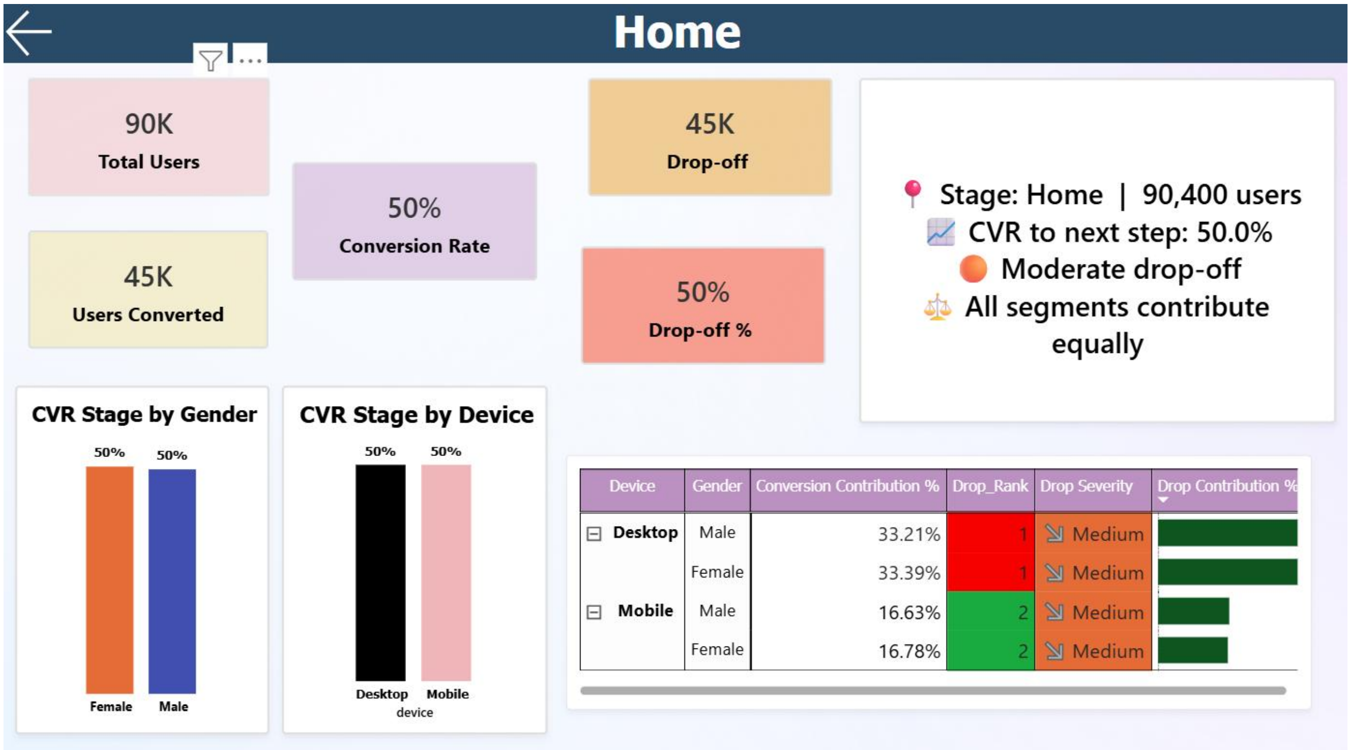
Stage Drill-Through Pages

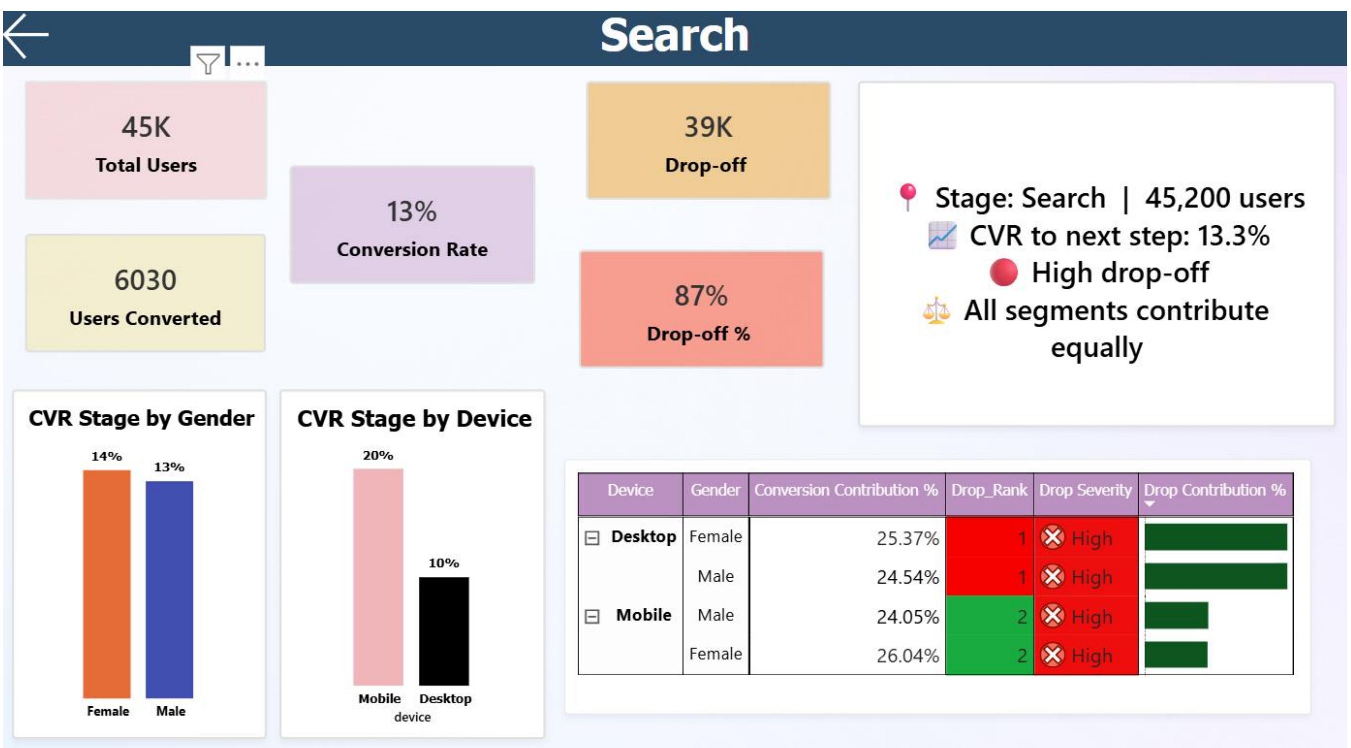
Click into any stage — visuals, KPIs, and insight text all update to the selected stage context

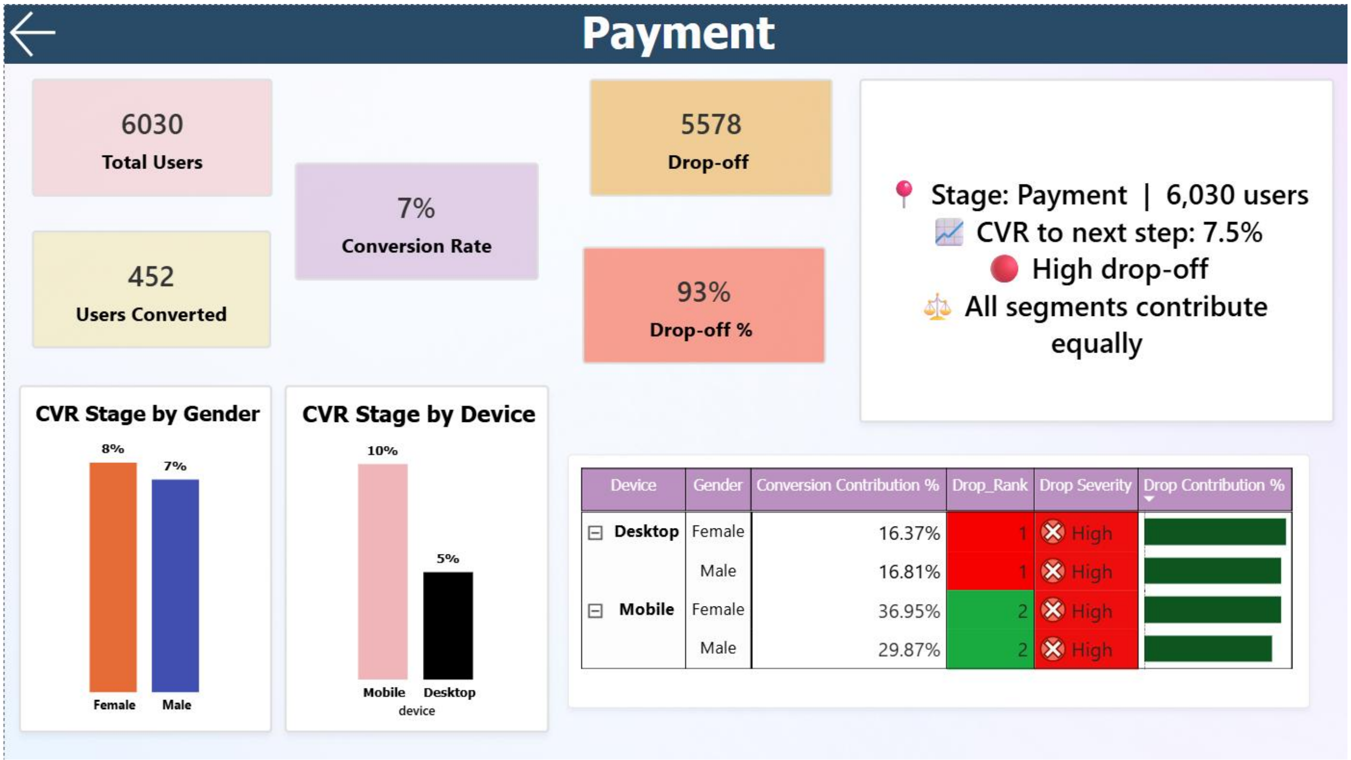
How It Works

Right-click any stage bar or funnel segment on the Overview page to navigate to the dedicated drill-through. All visuals, KPI cards, and the Dynamic Insight Text card update automatically. SELECTEDVALUE() in DAX reads the drill-through filter directly — no ALL() override needed.

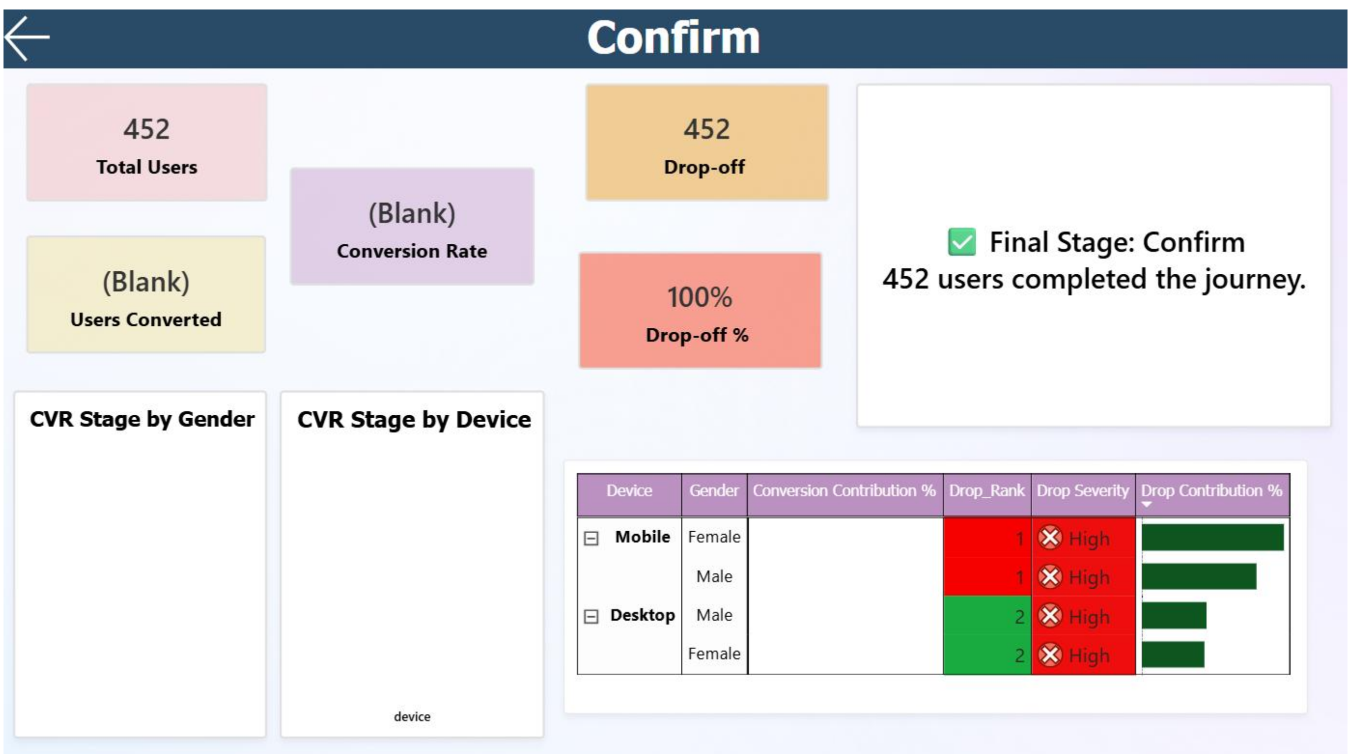
Stage	Users	Drop-off	Top Segment	Insight Card Output
Home	90,400	**50%**	All equal	Moderate drop-off · All segments equal · CVR 50%
Search	45,200	**87%**	Desktop Female 25.37%	High drop-off · Desktop Female leads · Critical stage
Payment	6,030	**93%**	Mobile Female 36.95%	High drop-off · Mobile Female leads · Highest severity
Confirm	452	—	—	Final stage · 452 users completed · Journey complete


Home
90K
Total Users
45K
Users Converted
50%
Conversion Rate
45K
Drop-off
50%
Drop-off %
Stage: Home | 90,400 users
CVR to next step: 50.0%
Moderate drop-off
All segments contribute equally
CVR Stage by Gender
50%
50%
Female
Male
CVR Stage by Device
50%
50%
Desktop
Mobile
device
Device
Gender
Conversion Contribution %
Drop_Rank
Drop Severity
Drop Contribution %
Desktop
Male
33.21%
1
Medium
Female
33.39%
1
Medium
Mobile
Male
16.63%
2
Medium
Female
16.78%
2
Medium


Search
45K
Total Users
6030
Users Converted
13%
Conversion Rate
39K
Drop-off
87%
Drop-off %
Stage: Search | 45,200 users
CVR to next step: 13.3%
High drop-off
All segments contribute equally
CVR Stage by Gender
14%
13%
Female
Male
CVR Stage by Device
20%
10%
Mobile
Desktop
device
Device
Gender
Conversion Contribution %
Drop_Rank
Drop Severity
Drop Contribution %
Desktop
Female
25.37%
1
High
Male
24.54%
1
High
Mobile
Male
24.05%
2
High
Female
26.04%
2
High


Payment
6030
Total Users
452
Users Converted
7%
Conversion Rate
5578
Drop-off
93%
Drop-off %
📍 Stage: Payment | 6,030 users
📈 CVR to next step: 7.5%
🔴 High drop-off
⚖️ All segments contribute equally
CVR Stage by Gender
8%
7%
Female
Male
CVR Stage by Device
10%
5%
Mobile
Desktop
device

Device	Gender	Conversion Contribution %	Drop_Rank	Drop Severity	Drop Contribution %
Desktop	Female	16.37%	1	High	
	Male	16.81%	1	High	
Mobile	Female	36.95%	2	High	
	Male	29.87%	2	High	


Confirm
452
Total Users
(Blank)
Users Converted
(Blank)
Conversion Rate
452
Drop-off
100%
Drop-off %
✅ Final Stage: Confirm
452 users completed the journey.
CVR Stage by Gender
CVR Stage by Device
device

Device	Gender	Conversion Contribution %	Drop_Rank	Drop Severity	Drop Contribution %
Mobile	Female		1	High	
	Male		1	High	
Desktop	Male		2	High	
	Female		2	High	

3. DAX Measures — A Glimpse

All calculations are explicit DAX measures — never implicit. This ensures correct filter context across drill-throughs, slicers, and cross-filtered visuals. Below is a quick tour of the key measures.

Core Measures

Stage Users

Counts users at the currently selected funnel stage. SELECTEDVALUE() ensures the measure reads from drill-through context, not a hardcoded filter.

```
Stage Users =
  CALCULATE(
    SUM(funnel_data[Users]),
    funnel_data[Stage] =
SELECTEDVALUE(funnel_data[Stage])
  )
```

CVR Stage

Conversion rate between consecutive stages. DIVIDE() returns 0 gracefully when no users exist at the denominator stage.

```
CVR Stage =
  DIVIDE([Stage Users Next],
           [Stage Users], 0)
```

Drop-off % Stage

The inverse of CVR — the proportion of users who did not advance to the next stage.

```
Drop-off % Stage =
  1 - [CVR Stage]
```

End-to-End CVR

Overall funnel conversion from Home to Confirm. The denominator is always pinned to Home-stage users — critical to prevent inflation from summing across all stages.

```
End-to-End CVR =  DIVIDE(
    CALCULATE(SUM(funnel_data[Users]), funnel_data[Stage] =
"Confirm"),
    CALCULATE(SUM(funnel_data[Users]), funnel_data[Stage] =
"Home") )
```

Segment & Classification Measures

Conversion Contribution %	Drop Rank	Drop Severity
Share of total conversions each device/gender segment owns. ALL() on the denominator removes segment filters so the ratio is against the full cohort.	Ranks device segments by users dropped — descending, DENSE. Surfaces which device group bleeds the most users at each stage.	Classifies drop-off: >60% = High · 40-60% = Medium · <40% = Low. Drives conditional formatting and the insight card's severity label.

Dynamic Insight Text — Drill-through Aware

The most technically rich measure in the dashboard. SELECTEDVALUE() reads the drill-through filter context, and the VAR block rebuilds the entire sentence structure depending on which stage is active:

```
Dynamic Insight =
VAR StageName   = SELECTEDVALUE(funnel_data[Stage])
VAR IsLastStage = StageName = "Confirm"
VAR DropLevel   = SWITCH(TRUE(),
    [Drop-off % Stage] > 0.6, "HIGH drop-off",
    [Drop-off % Stage] > 0.4, "MODERATE drop-off", "LOW drop-
off")
RETURN
  IF(IsLastStage,
    "Final Stage: " & StageName &
    FORMAT([Stage Users], "#,0") & " users completed the
journey.",
    "Stage: " & StageName & " | " & FORMAT([Stage Users],
"#,0") & " users" &
    "CVR to next: " & FORMAT([CVR Stage], "0.0%") &
    DropLevel &
    IF(IsEqual, "All segments equal", "Top: " & TopSeg)
  )
```

> **Key Technical Decision:** *SELECTEDVALUE() replaced ALL()/ALLSELECTED() in all drill-through measures. Without this, cards showed blended totals instead of the per-stage values the drill-through filter was supposed to isolate.*

4. Strategic Recommendations

Prioritized by estimated impact — calculated from users lost at each stage and realistic CVR uplift achievable through targeted UX or product changes.

#	Focus Area	Recommended Action	Impact
1	**Search → Payment**	Improve search relevance, add trust signals, simplify product discovery CTA	**High — largest volume gap**
2	**Desktop UX**	Simplify desktop checkout, reduce form fields, improve page load speed	**High — 50% of users**
3	**Payment → Confirm**	Add progress indicator, offer guest checkout, reduce friction steps	**Medium — high-intent users**
4	**March+ Trend**	Investigate pricing/platform changes correlating with converging drop-off	**Medium — trend risk**
5	**First-session Flow**	A/B test new-user onboarding to close gap between volume growth and CVR	**Long-term — scalable**

Skills Demonstrated

Advanced DAX	Data Modeling	Business Insight
SELECTEDVALUE · RANKX · SUMMARIZE · Dynamic text measures · Context-safe aggregation	Binary stage flags · Wide-to-long unpivot · Multi-table joins · Conditional funnel logic	Actionable recommendations mapped to measurable funnel outcomes with impact ranking